

CGM2IMG2Cal : A MultiModel network for prediction of lunch calories

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Abstract—This research presents a comprehensive analysis of Continuous Glucose Monitoring (CGM) data, focusing on preprocessing, feature engineering, and data augmentation techniques to derive meaningful insights into glycemic patterns. The study utilizes a dataset containing CGM readings, meal timings, and nutritional metadata. Key preprocessing steps include handling missing timestamps, imputing missing meal times with average values, and interpolating CGM data to ensure uniform time intervals. Feature engineering involves calculating metrics such as glucose variability, interquartile range (IQR), variance, time-to-peak, time-to-trough, and rates of glucose change. These features are critical for understanding the relationship between eating habits and glycemic responses. Advanced Python libraries such as Pandas, NumPy, and PyTorch are employed to build a scalable and efficient pipeline for data analysis. The study achieved an RMSE of 0.33 ± 0.01 , showcasing the model's effectiveness. The findings aim to support personalized nutrition strategies and diabetes management by providing a robust framework for analyzing CGM data. This work has potential applications in predictive health modeling and real-time monitoring systems for individuals with diabetes or those seeking to optimize their metabolic health.

I. INTRODUCTION

Continuous Glucose Monitoring (CGM) technology has revolutionized our understanding of glucose dynamics in the human body. This study delves into the intricate world of CGM data analysis, with the aim of uncovering patterns and insights that can transform personalized nutrition and diabetes management. The increasing prevalence of diabetes and metabolic disorders worldwide underscores the critical need for advanced tools to monitor and manage blood glucose levels. CGM devices provide a wealth of data, offering a continuous stream of glucose readings that capture the nuances of an individual's glycemic response to various stimuli, including food intake, physical activity, and stress. Our research leverages this rich dataset, combining CGM readings with detailed nutritional information, meal times, and a variety of metabolic markers. Recent advancements highlight the potential of CGM signals to predict meal macronutrient composition, offering automated insights into carbohydrate, protein, and fat content, which can significantly aid dietary monitoring and diabetes management (Huo et al., 2019) [1]. By integrating these diverse data sources, we aim to create a comprehensive picture of how different factors influence

glucose levels throughout the day. The potential applications of this work are far-reaching. From developing personalized dietary recommendations to improving predictive models for glucose management, the insights gained from this study could pave the way for more effective strategies to combat diabetes and promote general metabolic health. This introduction sets the stage for an in-depth exploration of CGM data analysis, highlighting its importance in the broader context of health and nutrition research.

II. OBSERVATIONS

A. Study Focus

The research will focus on preprocessing, feature engineering, and augmenting datasets such as CGM data, Viome data, Images, Demographics with meal and motion data from over 40 participants to build a multimodal model that can estimate lunch calories by incorporating meal photographs, micro gut health, sensing data, and breakfast information, and provide insights into glycemic patterns and personalized nutrition.

B. Dataset Overview

The dataset utilized in this study integrates multimodal data collected from multiple sources, including Continuous Glucose Monitoring (CGM) readings, Viome data, meal images, and participant demographics. This rich and diverse dataset is intended to explore the relationship between glycemic responses, microbiome health, nutrition, and individual characteristics, ultimately aiming to develop a personalized nutrition model.

a) Continuous Glucose Monitoring (CGM) Data: The CGM data forms the primary focus of this study, providing continuous, real-time measurements of blood glucose levels throughout the day. This time-series data, collected at frequent intervals (e.g., every 5 minutes), allows for the analysis of glucose fluctuations in response to different stimuli such as meals, physical activity, and sleep. The CGM dataset includes timestamps that denote the exact time each glucose measurement was recorded. This enables detailed exploration of glucose patterns, including glucose variability, time-to-peak, and time-to-trough metrics, all of which are crucial for understanding glycemic responses.

b) **Viome Data:** Viome data includes microbiome-related information collected through gut microbiome analysis, providing insights into the microbial composition of the gut. This data may include information about the abundance of various microbial species, gut health markers, and other related metrics. The relationship between gut health and glucose regulation has gained attention in recent research, and Viome data is instrumental in understanding how the microbiome influences metabolic health. By integrating this data with CGM readings, we can investigate the impact of gut microbiota composition on glucose fluctuations and metabolic responses to different meals and activities.

c) **Images of Meals:** The image dataset consists of photographs taken of the meals consumed by the participants. These photographs serve as a supplementary modality to understand meal composition and portion sizes. Image processing techniques are applied to analyze the content of the meal images, identifying food items and estimating nutritional values. By combining meal images with CGM and Viome data, we can explore how different food types and portion sizes affect blood glucose levels, offering potential insights into personalized dietary recommendations.

d) **Demographics Data:** The demographic data includes participant-specific information such as age, gender, health status, and other relevant characteristics. This data helps contextualize the glycemic responses and dietary effects observed in the study. For example, age and gender may influence metabolic responses to meals, while underlying health conditions like diabetes or insulin resistance may affect glucose regulation. The inclusion of demographic data allows for personalized analysis and the identification of subgroups within the dataset, which can be used to tailor nutritional strategies and health interventions.

e) **Data Collection Period:** The data spans multiple days, with some participants providing data over a period of up to 10 days, though not all participants have complete data for the entire duration. This longitudinal nature of the dataset allows for the analysis of glucose patterns over time, capturing the effects of various factors such as meal timings, nutrient composition, physical activity, and sleep on glucose regulation.

C. Units

- **CGM Data:** The glucose levels in the CGM data are primarily measured in *mg/dL* (or *mmol/L*).
- **Viome Data:** Viome data is expressed in *count-based metrics* for gut health.
- **Meal Images:** Meal images are digital and analyzed for *portion sizes* in *grams (g)* or *milliliters (mL)*.
- **Demographics:** Demographics are typically categorical, such as *age in years* and *gender*.
- **Motion Data:** Motion data is recorded in units such as *acceleration in m/s²*.
- **Data Collection Period:** The data collection period spans *days*, with time measurements in *minutes*.

III. METHODS

A. Data Preprocessing

The preprocessing phase for the entire dataset, encompassing Continuous Glucose Monitoring (CGM) data, Viome data, image data, and demographics, was conducted systematically to ensure consistency and quality across all modalities.

For the **CGM data**, preprocessing began with addressing missing values in the "Lunch Time" column by imputing the average lunch time (12:58 PM). The raw CGM data strings were converted into Python lists of tuples, each containing a timestamp and corresponding glucose reading. All timestamps were standardized to a uniform "HH:MM:SS" format to maintain consistency. Irregular intervals and missing readings were handled by interpolating CGM data at 5-minute intervals, ensuring a complete time series. This process involved identifying gaps larger than 5 minutes and filling them with values calculated based on adjacent readings. The analysis specifically focused on post-lunch data, beginning one hour after the meal, to capture the critical postprandial glucose response.

For the **Viome data**, raw comma-separated values were parsed into numerical values. Missing values were imputed using the mean for each feature. Each individual numerical value was treated as a standalone feature, while additional statistical features such as mean, standard deviation, minimum, and maximum were computed for each entry and incorporated into the dataset.

The preprocessing of **image data** involved converting image strings into tensors for computational processing. Rows containing empty or invalid image data were dropped to ensure the dataset contained only valid entries. To prepare the image data for analysis, pixel values were normalized, standardizing the inputs for subsequent modeling tasks.

For the **demographics data**, preprocessing involved handling categorical variables such as gender and property area through one-hot encoding to facilitate their integration with other data modalities. Continuous variables, such as age, were standardized to ensure uniform scales.

Finally, motion data, captured as part of the dataset, was processed to derive meaningful metrics such as steps taken, distance covered, or acceleration. All motion data was interpolated where gaps existed and aligned temporally with other data modalities.

These preprocessing steps ensured that each data type—CGM, Viome, images, demographics, and motion—was clean, consistent, and ready for integration into a multimodal analysis framework. This comprehensive preparation laid the groundwork for building robust predictive models that leverage the strengths of each modality.

B. Data Preparation

The dataset was meticulously processed and organized to ensure effective model training through a series of systematic steps. First, the CGM data, Viome data, demographic information, image data, and labels were integrated using Day

and Subject ID as the keys for merging. Following the integration, the Day and Subject ID columns, which were no longer required for model training, were dropped. The label column was then separated from the rest of the dataset, isolating the target variable.

Subsequently, the features were partitioned into distinct subsets for model training: x_{cgm} , containing the CGM data; x_{image} , comprising the image data related to lunch; and x_{neural} , which included the remaining structured features such as Viome data and demographics. This streamlined preprocessing ensured the dataset's readiness for robust and reliable analysis and modeling.

C. Held-Out Experiments

In this study, we conducted several experiments to assess the performance of different models for estimating the calories consumed during meals. The experiments involved utilizing various combinations of datasets, including Continuous Glucose Monitoring (CGM) data, Viome data, demographics, and meal images. Below are the results from the held-out experiments.

a) XGBoost Model with Viome Data, CGM Data, and Demographics: Initially, we used the XGBoost model with a combination of *Viome data*, *CGM data*, and *demographics*. The model achieved a Root Mean Squared Relative Error (RMSRE) value of **0.41**. This result demonstrated the potential of the model using these three modalities, but further improvements were necessary, so we continued to work on implementation of other models

b) Support Vector Regression (SVR) Model with CGM Data, Viome Data, and Demographics: Next, we implemented a *Support Vector Regression (SVR)* model using the same set of data: *CGM data*, *Viome data*, and *demographics*. The SVR model showed an improved performance, achieving a lower RMSRE value of **0.36**. This improvement indicated that the SVR model was better suited for the available data than the XGBoost model, although opened room for further enhancements and fine-tuning.

c) Multimodal Model with CGM Data, Viome Data, Demographics, and Images: Finally, we implemented a *multimodal model* that incorporated all the available data, including *CGM data*, *Viome data*, *demographics*, and *meal images*. This comprehensive model achieved the best performance, with an RMSRE value of **0.33**. The inclusion of images, along with the other datasets, led to a significant reduction in the RMSRE, demonstrating the effectiveness of combining multimodal data for more accurate predictions.

D. Implementation

We propose an Enhanced Multimodal Network designed to integrate and process multimodal data, including Continuous Glucose Monitoring (CGM) time-series data, neural feature data, and image data for predictive modeling. The architecture consists of three main subnetworks: CGM, neural, and image, followed by a fully connected layer that merges the extracted features to make predictions.

For the CGM subnetwork, two bidirectional Long Short-Term Memory (Bi-LSTM) layers are employed to capture both forward and backward temporal dependencies in the sequential data. The output h_t at each time step is passed through an attention mechanism, which computes the attention weight α_t to focus on the most relevant time steps. The attention-weighted outputs are then summed to obtain the final representation of the CGM data:

$$h_{\text{CGM}} = \sum_{t=1}^T \alpha_t h_t,$$

where $\alpha_t = \frac{\exp(q_t \cdot k_t)}{\sum_{i=1}^T \exp(q_i \cdot k_i)}$ represents the attention weight calculated using the query q_t and key k_t , derived from the LSTM outputs.

The neural subnetwork processes structured data, such as demographic features and viome data, including microbial counts. This data is passed through fully connected layers followed by ReLU activation functions to introduce non-linearity and improve model learning. The output h_{neural} from this subnetwork is given by:

$$h_{\text{neural}} = \text{ReLU}(W_{\text{neural}} X_{\text{neural}} + b_{\text{neural}}),$$

where W_{neural} and b_{neural} are the weights and biases of the fully connected layer. Dropout regularization is applied to reduce overfitting.

For the image subnetwork, a Convolutional Neural Network (CNN) is used to extract features from lunch-related images, such as food types and portion sizes, which are crucial for predicting calorie content. An attention mechanism is applied to focus on the most critical regions of the image, enhancing the model's ability to extract relevant information. The attention-weighted features are passed through dense layers for further refinement. The output of this process is expressed as:

$$h_{\text{image}} = \sum_{i=1}^N \alpha_i h_{\text{conv},i},$$

where α_i is the attention weight for the i -th convolutional feature map.

The outputs from the CGM, neural, and image subnetworks are then concatenated into a single feature vector $h_{\text{fusion}} = [h_{\text{CGM}}, h_{\text{neural}}, h_{\text{image}}]$ and passed through fully connected layers for final feature transformation and prediction. Dropout regularization is applied throughout the fully connected layers to prevent overfitting. The final output $\hat{y}$ is obtained through:

$$\hat{y} = W_{\text{output}} h_{\text{fusion}} + b_{\text{output}},$$

where W_{output} and b_{output} are the weights and bias of the output layer.

The model is trained using the Root Mean Squared Relative Error (RMSRE) loss function, which is defined as:

$$\text{RMSRE} = \sqrt{\frac{1}{N} \sum_{i=1}^N \left(\frac{|\hat{y}_i - y_i|}{y_i} \right)^2},$$

where $\hat{y}_i$ is the predicted value, y_i is the true value, and N is the number of samples.

The Adam optimizer is used for training, with the following update rule for the weights:

$$\theta_t = \theta_{t-1} - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t + \epsilon}},$$

where $\hat{m}_t$ and $\hat{v}_t$ are the biased-corrected first and second moments, η is the learning rate, and ϵ is a small constant added for numerical stability.

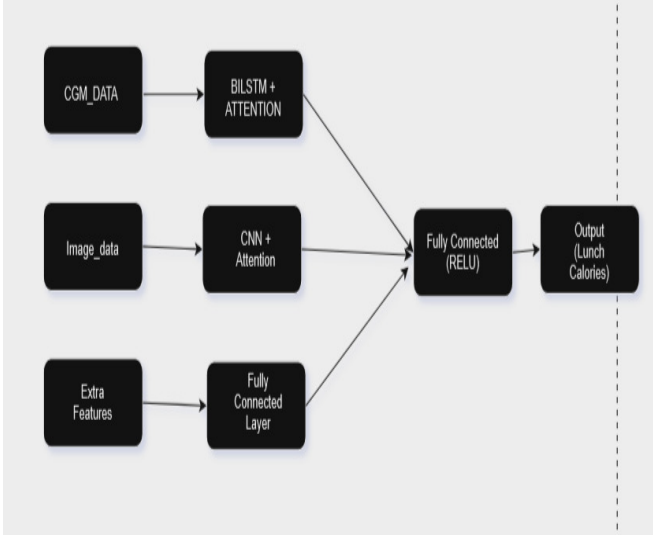


Fig. 1. Model Architecture

From Fig 1, this architecture effectively integrates temporal, structured, and unstructured data modalities to enhance model performance. The use of attention mechanisms ensures the model focuses on key information from each data source, improving predictive accuracy across multimodal data sources.

E. Result Analysis

We conducted a series of experiments to evaluate the performance of different models on predicting the calorie intake using multimodal data, including Continuous Glucose Monitoring (CGM) time series, Viome data, demographics, and images. The performance of the models was assessed using the Root Mean Squared Relative Error (RMSRE) metric, and the results are summarized as follows:

a) Initial Experiments with XGBoost: Initially, we employed the *XGBoost* model with a combination of *Viome data*, *CGM data*, and *demographics*. The model achieved a RMSRE value of **0.41**, providing an initial baseline for the predictive task. Despite its decent performance, there was room for improvement in incorporating temporal dependencies and multimodal features.

b) Improved Performance with Support Vector Regression (SVR): Next, we implemented a *Support Vector Regression (SVR)* model, using the same features: *CGM data*, *Viome data*, and *demographics*. The performance improved,

achieving a lower RMSRE value of **0.36**. This result showed the benefit of using SVR over XGBoost for this type of data, but further enhancement was still possible by including additional features, such as meal images.

c) Enhanced Performance with a Multimodal Model:

Finally, we designed and implemented a Multimodal Model, integrating all available data sources: CGM data, Viome data, demographics, and meal images. This model leveraged an Enhanced Multimodal Network, which combined three sub-networks for processing each modality. To further optimize the model's performance, we tuned the learning rate and weight decay parameters. Specifically, we tested the following values: Learning rates: 0.0001, 0.001, 0.01

Weight decays: 1e-4, 1e-5, 1e-3

Through hyperparameter tuning, we achieved the best performance with a learning rate of 0.0001 and a weight decay of 1e-4, resulting in an improved RMSRE value of 0.33. This significant reduction in RMSRE demonstrates the power of integrating multimodal data, as the model was able to leverage the complementary information provided by the different data types. The optimization of hyperparameters contributed to the model's improved accuracy, showcasing the importance of careful tuning for enhancing predictive performance.

d) Performance Summary: The TABLE 1 summarizes the RMSRE values achieved by the different models in our experiments:

Model	RMSRE Value
XGBoost (Viome, CGM, Demographics)	0.41
SVR (Viome, CGM, Demographics)	0.36
Multimodal Model (CGM, Viome, Demographics, Images)	0.33

TABLE I
PERFORMANCE COMPARISON OF DIFFERENT MODELS

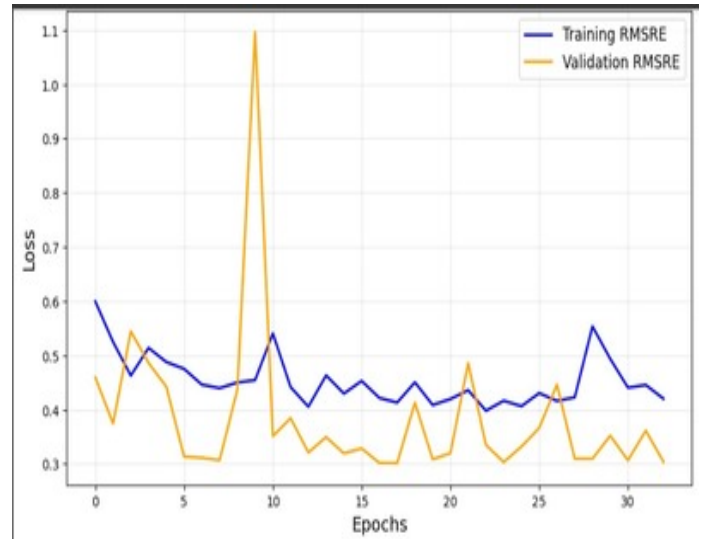


Fig. 2. Training vs Validation RMSE

From Fig 2, we can infer the values of RMSE on training and validation sets across epochs (considering epochs on x-axis and loss on the y-axis). The multimodal model demonstrates

superior performance compared to the XGBoost and SVR models, achieving the lowest RMSRE value of **0.33**. This improvement underscores the power of leveraging diverse data sources in the prediction process. By incorporating image data alongside CGM, Viome, and demographic features, the model effectively captures comprehensive and complementary information, leading to enhanced prediction accuracy.

The superior performance of the multimodal model can be attributed to the processing of CGM data through LSTM layers, which effectively capture temporal patterns in the time-series data. These temporal features are further enriched by CNN-extracted features from image data. This integration allows the model to synergistically learn from both CGM and image data, in addition to other structured features, enabling it to better understand complex relationships within the data.

The inclusion of image data is particularly impactful, as it provides critical insights into meal portion sizes, composition, and other factors influencing glycemic response. The significant reduction in RMSRE from **0.36** with the SVR model to **0.33** with the multimodal model highlights the importance of integrating all available data modalities. This holistic approach ensures a more robust and accurate prediction process.

IV. CONTRIBUTIONS

Each team member in this research independently contributed by implementing a distinct model, with data preprocessing tailored to meet the specific requirements of their respective datasets. The individual contributions are as follows:

- **Sai Jaideep Reddy Mure:** Managed the preprocessing and modeling of CGM data. He was responsible for drafting the Abstract, Introduction, Methods (Data Preprocessing), and compiling the References for the report.
- **Hitha MV:** Focused on processing image data and developing its corresponding model. She contributed to the Methods (Data Preprocessing, Data Preparation, Implementation) and the Conclusion sections of the report.
- **Phani Jyothi Kurada:** Handled the Viome and demographic data, building a model tailored to these structured datasets. She wrote the sections on Observations, Methods (Heldout Experiments, Result Analysis), and the Contributions section in the report.

In the final phase, the team collaboratively combined their efforts, rigorously testing and fine-tuning the models with various hyperparameters. This joint effort culminated in the development of a high-performing multimodal model that integrated the strengths of all individual components.

V. CONCLUSION

This study underscores the power of multimodal data in improving calorie prediction accuracy. Initially, models utilizing CGM, Viome, and demographic data showed moderate performance, with a notable improvement when applying support vector regression. However, the integration of meal images alongside these datasets led to the most significant enhancement, demonstrating that combining diverse data sources, including temporal, structured, and unstructured information,

can substantially boost predictive performance. These results emphasize the potential of multimodal approaches for advancing personalized nutrition and meal estimation models.

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